

P O R

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 60 / Pass Marks: 24

MCA103/MCA113: Operating System (New/Old Course)

Candidates are required to give their answers in their own words as far as practicable.

Group A

Answer TWO questions.

2×12=24

1. Define process scheduling and briefly explain the performance criteria for project scheduling. Explain preemptive and non-preemptive shortest-job first (SJF) scheduling with suitable example. (6+6)
2. Why is memory management necessary? Discuss about fixed partitioning and dynamic partitioning, with suitable examples, and also mention their advantages and disadvantages. 2+10
3. What is Resource Allocation Graph (RAG)? Explain the conditions for deadlock. Determine if the system is safe from deadlock or not, using Banker's Algorithm for given condition. [2+4+6]

Request Matrix			
Process	A	B	C
P1	11	6	4
P2	3	4	1
P3	8	2	3
P4	3	2	4
P5	5	2	3

Allocation Matrix			
Process	A	B	C
P1	3	2	1
P2	3	1	0
P3	4	1	1
P4	1	0	1
P5	2	0	1

MAX Resource Matrix		
A	B	C
17	8	10

Group B

Answer SIX questions.

6×6=36

4. What is IPC? Explain process states with neat and clean diagram. 1+5
5. What are different attributes of file? Explain different operations of directories. 2+4
6. Differentiate between paging and segmentation, with examples.
7. Does DMA increase performance of system? Explain its working phenomena. (1+5)

Contd. ...

(2)

8. ✓ A disk queue has the following request to read tracks: 87, 170, 4, 57, 173, 32, 67, 183. If disk head is initially at cylinder 90 and there are total 200 tracks, then calculate total head movement using FCFS, SSTF and SCAN algorithms. Assume that read/write head was moved to 63 from 90 initially. 713
265
364
9. ✓ "Increasing the frame size, number of page fault will decrease". With proper example, explain if the given statement is valid or not.
10. ✓ Explain client-server computing model of distributed systems.
11. ✓ Write short notes on Any TWO:
- (a) Trusted system
 - (b) Semaphore
 - (c) History of operating system

name
identifier
size
location
type

positive integer